

Linear regression (simple and inference)

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Simple Linear Regression: The Foundations

Statistical Foundations

Inference in Simple Linear Regression



What lifestyle factors do you think are predictive of body fat percentage?



Problem: Predict body fat percentage based on observed lifestyle characteristics

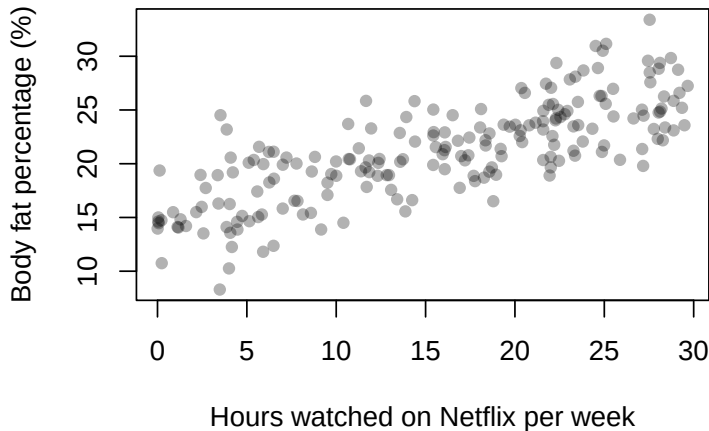
Solution:

- Look at data where we know body fat percentage and some observed characteristics
- Build a decision rule that predicts body fat percentage as a function of the observed characteristics

To keep things simple, let's focus only on **hours watched on Netflix per week**.



- The value we seek to predict is the **response** (or dependent/output) variable:
 - Y = body fat percentage (%)
- The variable we use to guide prediction is the **predictor** (or explanatory/input) variable:
 - X = hours watched on Netflix per week



View the data as points in the $X \times Y$ plane. There seems to be a **linear** relationship!



Y = response or outcome variable

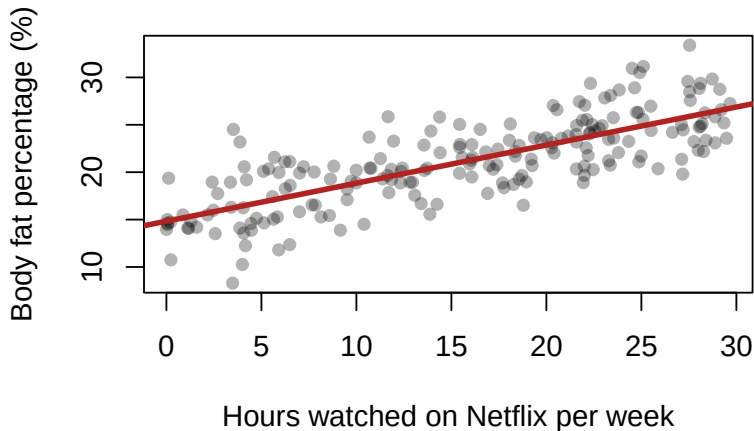
X = explanatory or predictor variable

A **linear** relationship is written:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- β_0 is the **intercept** (in units of Y)
- β_1 is the **slope** (in units of Y per unit of X)
- ε is the **error** or residual

Recall the equation of a line: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$



As Netflix hours go up, body fat percentage tends to go up. But how do we find the “best” line?



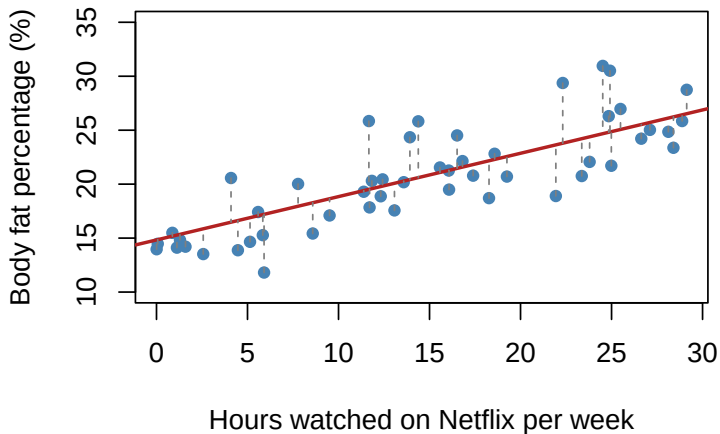
- The **fitted value** $\hat{Y}_i$ is our prediction for observation i :

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

- The **residual** e_i is how wrong our prediction is:

$$e_i = Y_i - \hat{Y}_i$$

- We can write: $Y_i = \hat{Y}_i + e_i$ (observed = predicted + error)



The dashed lines are the residuals: $e_i = Y_i - \hat{Y}_i$



How do we find the “best” line?

Ideally, we want to minimize the size of all residuals:

- If they were all zero, we'd have a perfect fit
- But there's a trade-off: moving closer to some points means moving away from others



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Least Squares: Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize $\sum_{i=1}^n e_i^2$

$$\sum_{i=1}^n e_i^2 = (Y_1 - \hat{Y}_1)^2 + (Y_2 - \hat{Y}_2)^2 + \cdots + (Y_n - \hat{Y}_n)^2$$



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- Leads to nice mathematical properties (unique solution)
- The formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ are:

$$\hat{\beta}_1 = r_{xy} \times \frac{s_y}{s_x}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

where r_{xy} is the correlation and s_x , s_y are standard deviations

```
model <- lm(Body_Fat ~ Netflix_Hours, data=netflix_data)
summary(model)
```

```
Call:
lm(formula = Body_Fat ~ Netflix_Hours, data = netflix_data)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-7.9344 -1.8695  0.0093  1.8318  8.2710
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  14.82928    0.41936   35.36  <2e-16 ***
Netflix_Hours  0.40153    0.02338   17.18  <2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 2.889 on 198 degrees of freedom
Multiple R-squared:  0.5984, Adjusted R-squared:  0.5964
F-statistic: 295 on 1 and 198 DF, p-value: < 2.2e-16
```



Our prediction equation is:

$$\widehat{\text{Body Fat}} = 14.83 + 0.4 \times \text{Netflix Hours}$$

- **Intercept** ($\hat{\beta}_0 = 14.83$): Predicted body fat percentage when Netflix Hours = 0.



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- **Intercept** ($\hat{\beta}_0 = 14.83$): Predicted body fat percentage when Netflix Hours = 0.
- **Slope** ($\hat{\beta}_1 = 0.4$): Each additional hour of Netflix per week is associated with a 0.4 percentage point increase in predicted body fat.
- A 10 hour increase $\rightarrow$ 4 percentage point increase in predicted body fat

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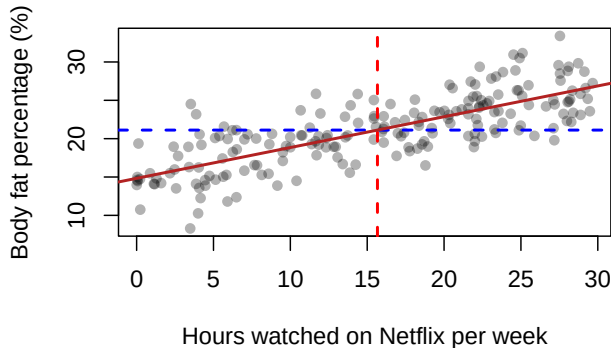


To understand how we get the slope and intercept, we need to understand how X and Y vary together.

- **Variance** measures how one variable spreads out: $s_y^2 = \frac{1}{n-1} \sum (Y_i - \bar{Y})^2$
- **Covariance** measures how two variables vary *together*:

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

- Positive covariance: When X is above its mean, Y tends to be above its mean
- Negative covariance: When X is above its mean, Y tends to be below its mean
- Zero covariance: No linear relationship



The covariance measures how points fall relative to the mean lines. Points in quadrants I and III contribute positively; points in II and IV contribute negatively.



Covariance is hard to interpret because it depends on the units of measurement.

Correlation standardizes the covariance:

$$r = \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{s_x s_y} = \frac{\text{cov}(X, Y)}{\sqrt{s_x^2 s_y^2}}$$



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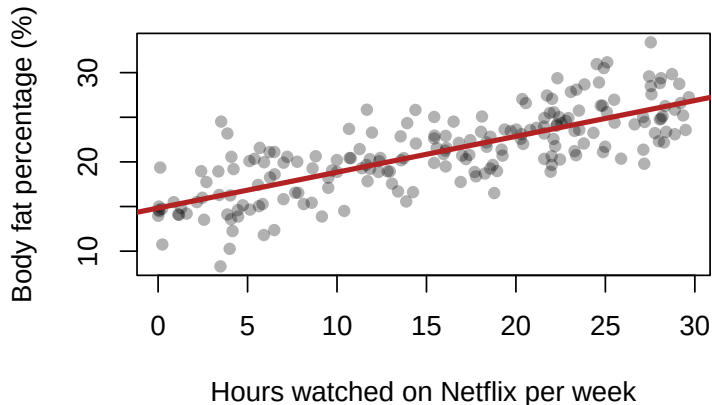
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- $r > 0$: positive linear relationship
- $r < 0$: negative linear relationship
- $r = 0$: no linear relationship
- $|r| = 1$: perfect linear relationship

Correlation for our Netflix example



```
cor(netflix_data$Netflix_Hours, netflix_data$Body_Fat)
```

```
[1] 0.7735597
```



How do we interpret this correlation?



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- If correlation is zero, the slope is zero (no linear relationship)
- The intercept ensures the regression line passes through $(\bar{X}, \bar{Y})$



Two key questions:

1. How much of the variation in Y does our model explain?
2. How wrong are our predictions, on average?



Remember: $Y = \hat{Y} + e$ (observed = predicted + residual)

The total variation in Y can be split:

$$\underbrace{\sum (Y_i - \bar{Y})^2}_{\text{SST (Total)}} = \underbrace{\sum (\hat{Y}_i - \bar{Y})^2}_{\text{SSR (Regression)}} + \underbrace{\sum e_i^2}_{\text{SSE (Error)}}$$

- **SST**: Total variation in Y
- **SSR**: Variation explained by the regression
- **SSE**: Variation left unexplained (residuals)



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- About 59.8% of the variation in body fat percentage is explained by Netflix hours
- Note: For simple regression, $R^2 = r^2$ (correlation squared)



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- Interpretation: The typical size of our prediction errors
- For our model: $s_e = 2.89$
- On average, our predictions are off by about 2.89 percentage points
- Why $n - 2$? We “used up” 2 degrees of freedom estimating $\hat{\beta}_0$ and $\hat{\beta}_1$



The **simple linear regression model** assumes:

$$Y = \beta_0 + \beta_1 X + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

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- This means: $Y|X \sim N(\beta_0 + \beta_1 X, \sigma^2)$



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Let's see these ideas in action with a new example!

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What personal characteristics about an instructor do you think are predictive of the scores they receive on student evaluations?



Economics of Education Review

Volume 24, Issue 4, August 2005, Pages 369–376

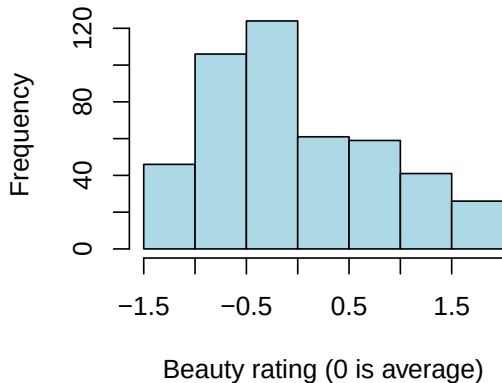
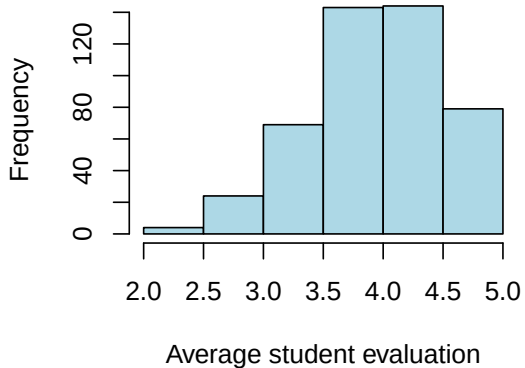


Beauty in the classroom: instructors' pulchritude and putative pedagogical productivity

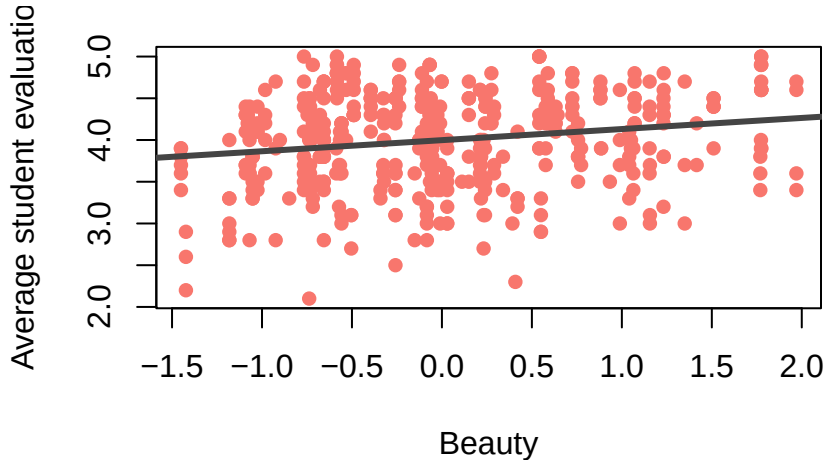
Daniel S. Hamermesh  , Amy Parker



- Student evaluations of $N = 463$ instructors at UT Austin, 2000-2002
- For each instructor:
 - **eval**: average student evaluation of teacher
 - **beauty**: average beauty score from a six-student panel
 - **gender**: male or female
 - **credits**: single- or multi-credit course
 - **age**: age of instructor
 - (and more...)



Do you think there is a positive or negative relationship between beauty and teaching ratings?



```
cor(profs$eval, profs$beauty)
```

```
[1] 0.189
```

Let's build a simple regression model



```
model <- lm(eval ~ beauty, data=profs)
summary(model)
```

```
Call:
lm(formula = eval ~ beauty, data = profs)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.8002 -0.3630  0.0725  0.4021  1.1037
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.9983     0.0253   157.73 < 2e-16 ***
beauty         0.1330     0.0322    4.13 4.2e-05 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.545 on 461 degrees of freedom
Multiple R-squared:  0.0357, Adjusted R-squared:  0.0336
F-statistic: 17.1 on 1 and 461 DF,  p-value: 4.25e-05
```



- `eval` is the **response** variable (Y); `beauty` is the **predictor** variable (X)



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$$\widehat{\text{eval}} = 3.998 + 0.133 \cdot \text{beauty}$$



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- We can predict the evaluation score for someone based on their beauty score. What is the predicted evaluation score for someone with a beauty score of 1?

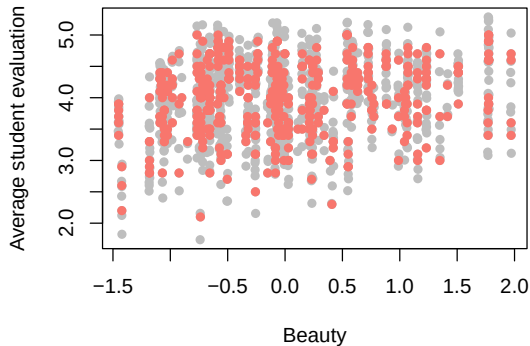


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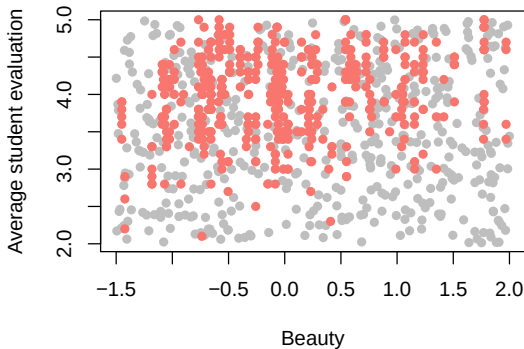
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- What do the coefficients mean?

Is this the population?

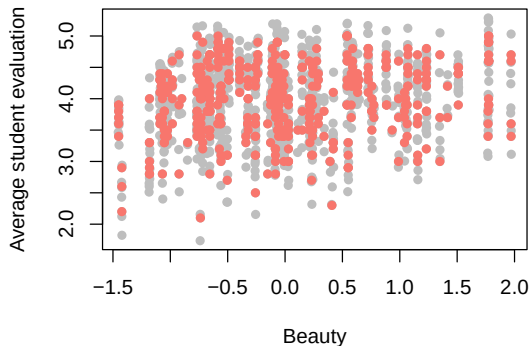


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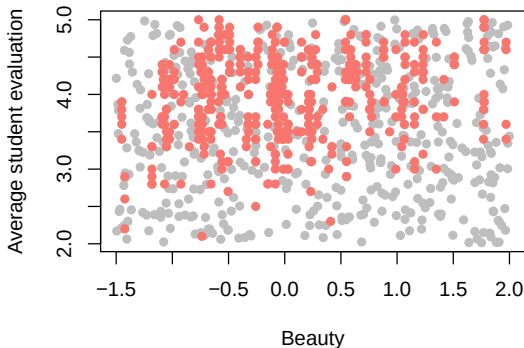


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Or is this the population?



- The population line $Y = \beta_0 + \beta_1 X$ is what we actually care about, and we can never see it
- $\hat{\beta}_0$ and $\hat{\beta}_1$ are our best estimates of it. Inference is about how close they are.



	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.9983	0.0253	157.73	< 2e-16 ***
beauty	0.1330	0.0322	4.13	4.2e-05 ***

- 95% CI for the slope is $\hat{\beta}_1 \pm t^*SE(\beta_1)$ (estimate $\pm$ (critical value)(standard error))



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- t^* is the critical value from a t distribution with $n - 2$ degrees of freedom: $\text{qt}(0.975, 463-2) \approx 1.97$



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- **Rule of thumb** for anything in this class: (estimate) ± 2 (standard errors)
- The standard error measures how far our *estimate* is likely to be from the *population value*

```
confint(model)
```

	2.5 %	97.5 %
(Intercept)	3.9485	4.048
beauty	0.0698	0.196

- β_1 : We are 95% confident that the incremental impact of each additional beauty point is between 0.07 and 0.20 student evaluation points.

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```

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(Intercept)	3.9485	4.048
beauty	0.0698	0.196

- β_1 : We are 95% confident that the incremental impact of each additional beauty point is between 0.07 and 0.20 student evaluation points.
- β_0 : We are 95% confident that the average student evaluation score for average-looking professors (beauty = 0) is between 3.95 and 4.05.



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95% confidence intervals for predictions (aka prediction intervals): The interval captures 95% of *future y values* for a given x

Testing whether the slope is really zero



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beauty	0.1330	0.0322	4.13	4.2e-05 ***

- If $\beta_1 = 0$, then beauty has no impact on eval in the population

Testing whether the slope is really zero



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- If $\beta_1 = 0$, then beauty has no impact on eval in the population
- $\hat{\beta}_1 = 0.133 \neq 0$ in our sample, but was that just due to chance?

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- Test the null $\beta_1 = 0$ against the alternative $\beta_1 \neq 0$, accepting a false positive rate (type I error) of $\alpha = 0.05$

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- Test the null $\beta_1 = 0$ against the alternative $\beta_1 \neq 0$, accepting a false positive rate (type I error) of $\alpha = 0.05$
- $\alpha = 0.05$ means we accept a 5% chance of rejecting a true null. You must pick it before looking at the data.



	Estimate	Std. Error	t value	Pr(> t)
beauty	0.1330	0.0322	4.13	4.2e-05 ***

- The p -value for the slope is 4.2×10^{-5} ; since it is very small, it is very unlikely that $\hat{\beta}_1$ is this big just by chance if β_1 is truly zero



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- The p -value is *not* the probability that the null is true. It *assumes* the null and measures how unlikely our result would be by chance
- Connection to CIs: a 95% CI contains *all the values* the test fails to reject at $\alpha = 0.05$, i.e. a **set of plausible values** your data doesn't rule out

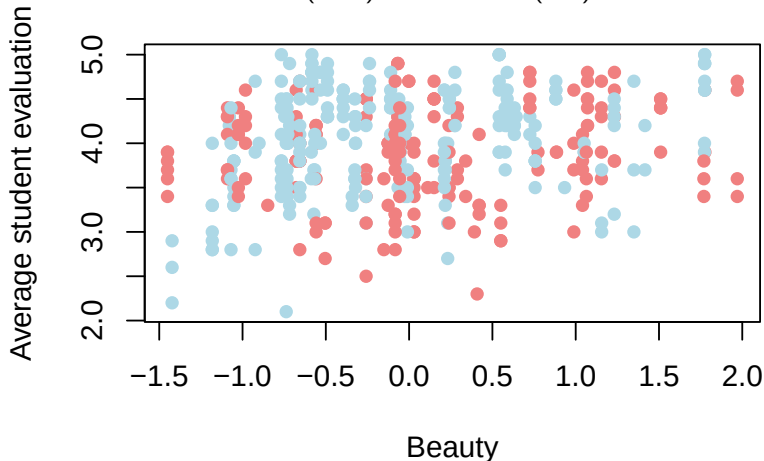
Is this the end of the story?

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(Spoiler alert: No.)

What's the impact of gender on student evaluations?



Do you see a difference between men (blue) and women (red)?



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